

Trainings ▾

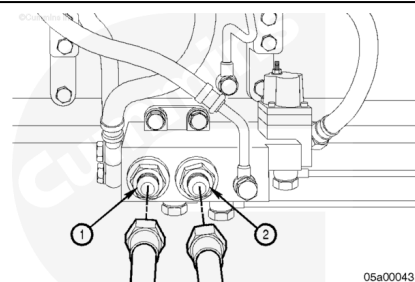
Remove

Low Pressure Fuel Line(s)

For generator and generator drive engines with mechanical lift pumps, the following steps **must** be completed:

Note : Thoroughly clean the area around the fuel lines before removal. Sketch or photograph the fuel line arrangement before removal. Tag the lines to identify them for assembly purposes.

Remove the tank fuel supply line (1) and the return to tank line (2).



For generator and generator drive engines with mechanical lift pumps, the following steps **must** be completed:

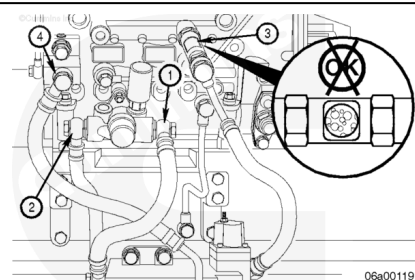
Remove the lift pump outlet line (1).

Remove the lift pump inlet line (2).

Remove the injection pump inlet line (3).

Remove the injection pump overflow line (4).

Refer to Procedure 006-013 in Section 6
(</qs3/pubsys2/xml/en/procedures/57/57-006-013.html>) to inspect low pressure and fuel drain lines for reuse.



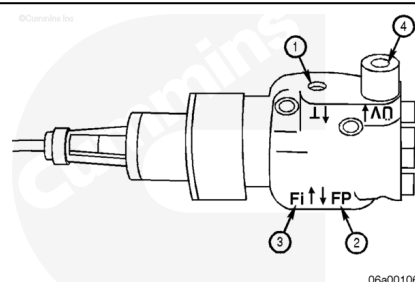
For industrial engines with mechanical lift pumps and an EHAB (fuel shutoff valve) device, the following steps **must** be completed:

Remove the EHAB supply line (1).

Remove the EHAB outlet line (2).

Remove the EHAB injection pump inlet line (3).

Remove the EHAB injection pump overflow line (4).



For industrial engines with mechanical lift pumps and an EHAB (fuel shutoff valve) device, the following steps **must** be completed:

Disconnect the low-pressure lines from the fuel manifold blocks.

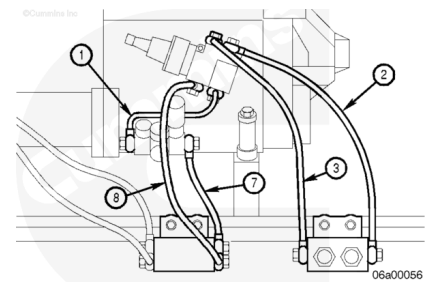
Remove the mechanical lift pump supply line (1).

Remove the injection pump overflow line (2).

Remove the fuel supply to EHAB line (3).

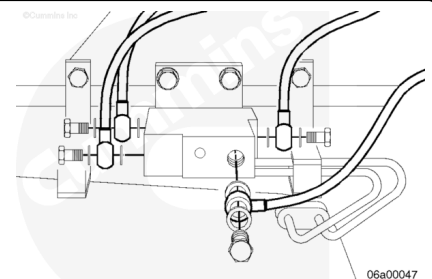
Remove the lift pump outlet line (7).

Remove the fuel injection pump inlet line (8).



Note : Thoroughly clean the area around the fuel lines before removal. Sketch or photograph the fuel line arrangement before removal. Tag the lines to identify them for assembly purposes.

For industrial engines with mechanical lift pumps, remove the low-pressure fuel lines from the mechanical fuel lift pump and the fuel filter head.



For industrial engines and generator/generator drive engines with electric lift pumps, the following steps **must** be followed to remove the low-pressure fuel lines from the system.

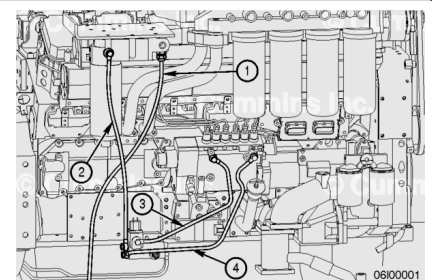
Note : Thoroughly clean the area around the fuel lines before removal. Sketch or photograph the fuel line arrangement before removal. Tag the lines to identify them for assembly purposes.

Remove the lift pump fuel supply line (1).

Remove the fuel line (2) from the filter to the fuel manifold.

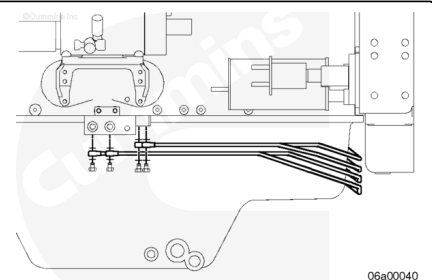
Remove the injection pump inlet line (3).

Remove the injection pump overflow line (4).



Remove the fuel crossover lines from the left bank fuel connection block, or manifold, and the right bank fuel connection block, or manifold.

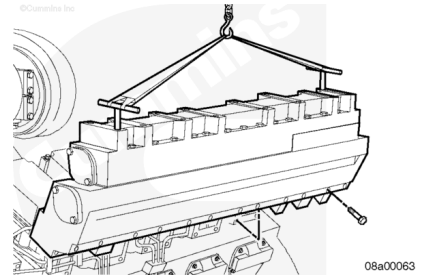
Refer to Procedure 006-021 in Section 6 (</qs3/pubsys2/xml/en/procedures/57/57-006-021-tr.html>) for removal and installation of fuel drain manifold on high pressure fuel pump and fuel drain lines from rocker lever housings.



High Pressure Fuel Line(s)

Remove the aftercooler assembly. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/57/57-010-002-tr.html>)

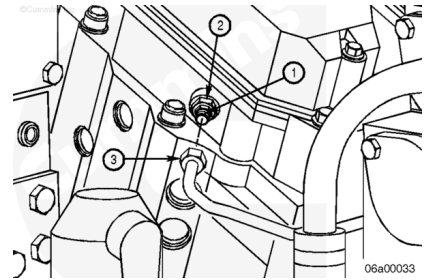
Note : The aftercooler assembly **must** be removed if high- pressure injector supply lines are to be removed.



⚠ CAUTION ⚠

The high pressure fuel supply lines are shaped to provide fuel to the injectors at the proper point in the cycle. Changes to the shape or dimensions of these lines will affect timing.

Do not bend the high pressure fuel supply lines. Engine performance can be impaired or damage can occur.



Note : Thoroughly clean the area around the fuel lines before removal.

Note : Mark or photograph all fuel tubing, mounting pads, and brackets for identification purposes, to aid in reassembly.

Disconnect the high-pressure injector fuel supply lines (3).

Loosen the locknut (2) and remove the connector (1).

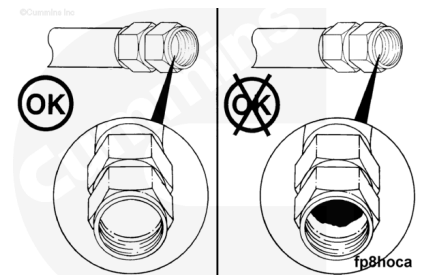
Clean and Inspect for Reuse

Note : The procedure that follows is for rubber or braided hoses.

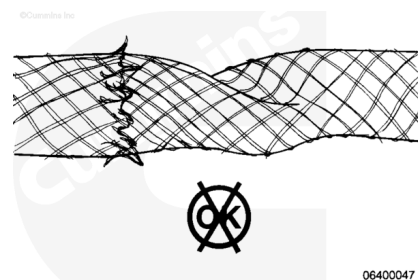
Inspect the inside of the hose from the fuel tank to the fuel filter for any separation of the liner.

Inspect for any pinches or loops in the hose that could obstruct the flow.

If damage is found, the part **must** be replaced.

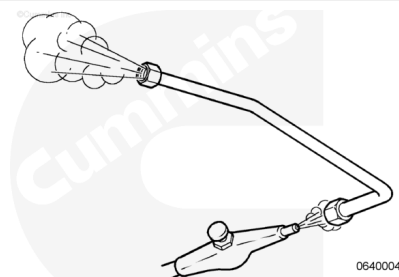


For the braided fuel supply lines, inspect the hoses for frayed mesh, and pinched areas that prevent the proper fuel flow. The hose **must** be replaced if it is damaged.



⚠ WARNING ⚠

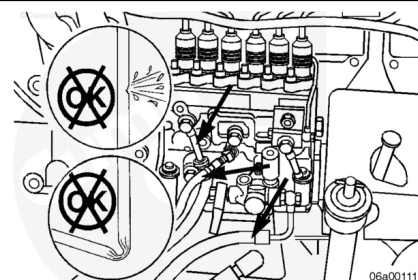
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



For rigid steel fuel supply lines, use compressed air to flush the lines and remove any loose dirt particles.

⚠ CAUTION ⚠

The high-pressure fuel supply lines are shaped to provide fuel to the injectors at the proper point in the cycle. Changes to the shape or dimensions of these lines will affect timing. Do not bend the high-pressure fuel supply lines. Engine performance can be impaired or damage can occur.



Inspect the fuel lines for cracks that can cause a loss of pressure.

Inspect the fuel lines for sharp bends that can cause a restriction in pressure.

The part **must** be replaced if damaged is found.

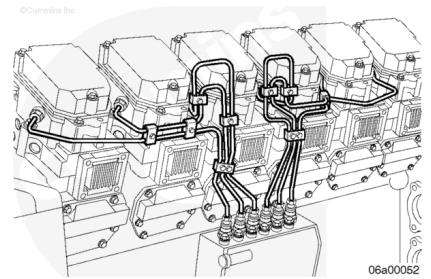
Install

High Pressure Fuel Line(s)

⚠ CAUTION ⚠

When installing flexible hoses, make sure that all the original clips are fitted and that the hoses do not contact any other hoses or engine components. Interference between hoses

and other engine components can lead to leakages and engine damage.



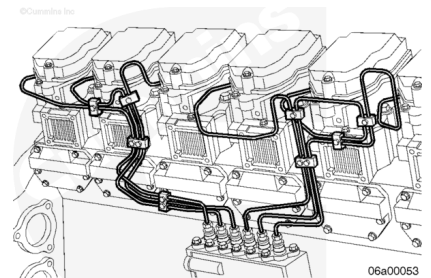
Note : Make sure there is a minimum of 15 mm (5/8 in) between flexible hoses and hard lines.

Install the left bank high-pressure fuel lines.

Torque Value: 23 n•m [204 in-lb]

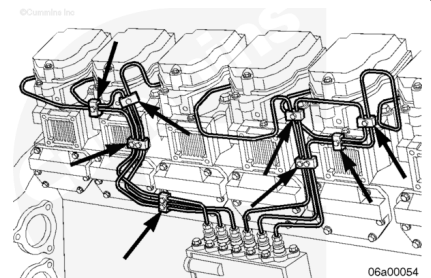
Install the right bank high-pressure fuel lines.

Torque Value: 23 n•m [204 in-lb]



⚠ WARNING ⚠

High-pressure fuel line clamps must be installed as shown. If the clamps are not installed, the lines can break causing a fuel leak.



Install the tube clamp brackets onto the right bank of the engine.

Note : The tube clamp brackets are attached on the right bank between cylinders number 2 and number 3 and between cylinders number 4 and number 5, and by the number 4 cam follower cover.

Torque Value: 65 n•m [48 ft-lb]

Install the tube clamps to the tubes.

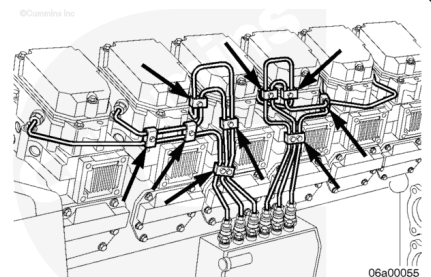
Note : The right bank uses five clamps.

Torque Value: 9 n•m [80 in-lb]

Note : The tube clamp with the spacer always goes with the turbocharger oil drain tube.

⚠ WARNING ⚠

High-pressure fuel line clamps must be installed as shown. If the clamps are not installed, the lines can break, causing a fuel leak.



Install the tube clamp brackets onto the left bank of the engine.

Torque Value: 65 n•m [48 ft-lb]

Install the tube clamps to the tubes.

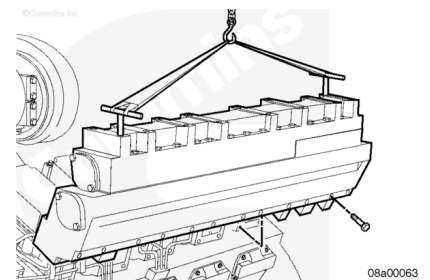
Torque Value: 9 n•m [80 in-lb]

Note : Left bank tube clamp brackets are attached between cylinders number 2 and number 3 and between cylinders number 3 and number 4.

Note : The left bank uses seven clamps.

Note : Some of the tube brackets are installed on top of the turbocharger oil drain tube flanges.

Install the aftercooler assembly. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/57/57-010-002-tr.html)



Low Pressure Fuel Line(s)

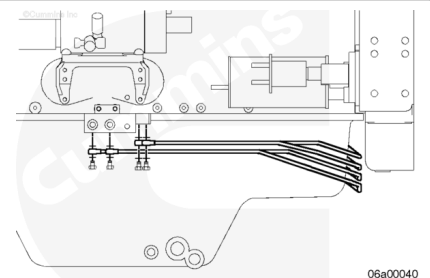
Install the low-pressure fuel crossover tubes to the left bank fuel connection block and to the right bank fuel connection block.

Note : All fuel lines and tubes have two sealing washers per banjo capscrew.

Torque Value: 27 n•m [239 in-lb]

Install the tube clamps.

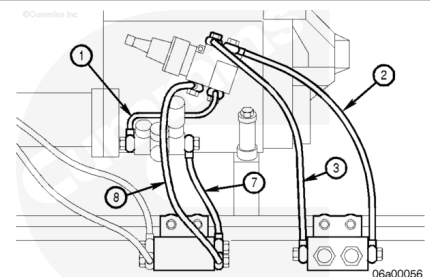
Torque Value: 9 n•m [80 in-lb]



For industrial engines with mechanical lift pumps and EHAB devices, the following low pressure fuel lines **must** be installed.

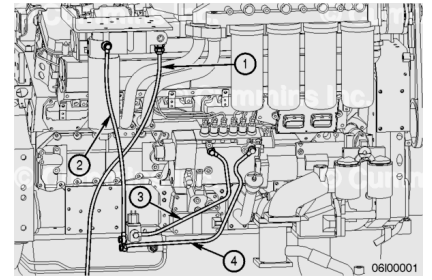
- EHAB supply (3)
- Lift pump supply (1)
- Fuel filter supply (7)
- EHAB supply filtered (8)
- Overflow (2).

Torque Value: 27 n•m [239 in-lb]



For industrial and generator/generator drive engines with electric left pumps, the following low-pressure fuel lines **must** be installed:

- Lift pump supply (1)
- Fuel manifold supply, filtered (2)
- Fuel injection pump supply (3)
- Overflow (4).



Hand tighten all connections.

Note : Use two sealing washers with each banjo connector around the fuel injection pump and the fuel manifold. Two different sizes of banjo connectors are used. Each requires a different torque value.

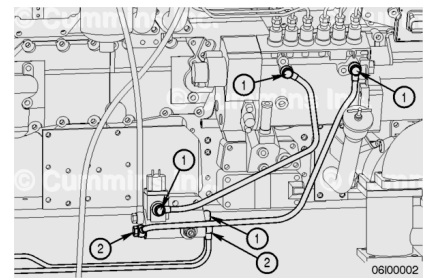
For industrial and generator/generator drive engines with electric lift pumps, apply the final torque to each banjo and line as indicated.

Use a 22 mm socket to tighten the banjo connectors at location (1).

Torque Value: 30 n•m [22 ft-lb]

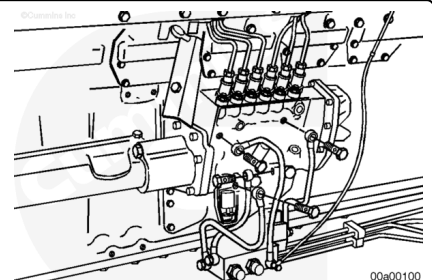
Use a 19 mm socket to tighten the banjo connectors at location (2).

Torque Value: 20 n•m [177 in-lb]



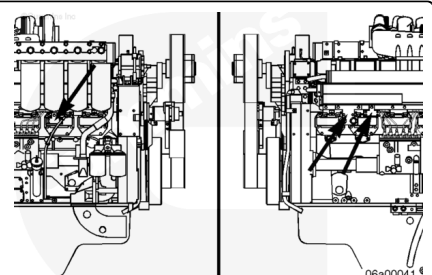
Install the following low-pressure fuel lines for a generator or generator drive engine with mechanical lift pumps.

- Install the fuel shutoff supply line.
- Install the fuel strainer supply line.
- Install the lift pump supply line.
- Install the fuel filter supply line.
- Install the injection pump supply line.
- Install the overflow line.



Torque Value: 27 n•m [239 in-lb]

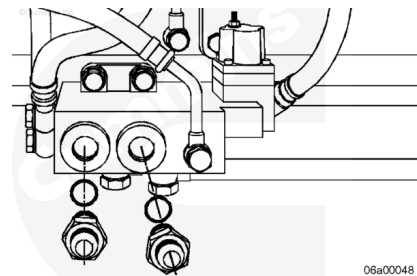
Install the engine harness and the brackets. Reference the following procedure in the QST30 Industrial Engine Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666214. Refer to Procedure 019-032 in Section 19 (</qs3/pubsys2/xml/en/procedures/87/87-019-032.html>)



For the fuel inlet line, install new o-rings on the fuel manifold inlet fitting (from tank) and fuel manifold outlet (return-to-tank) fitting.

Inspect for proper o-ring installation.

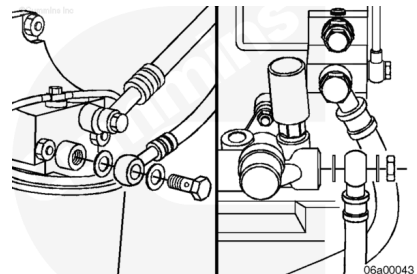
Torque Value: 80 n•m [59 ft-lb]



Install the hose at the fuel pump inlet and fuel filter head outlet fitting.

Tighten the hose.

Torque Value: 30 n•m [22 ft-lb]

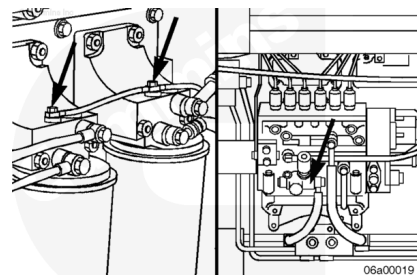


Vent

Note : This section applies **only** to engines with mechanical lift pumps.

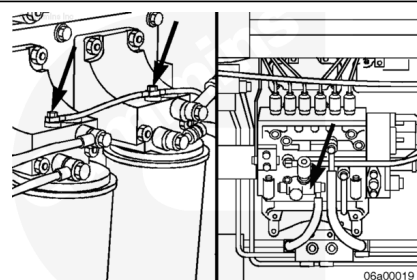
Controlled venting is provided at the injection pump through the fuel drain manifold. Small amounts of air introduced by changing the fuel filters or fuel injection pump supply line will be vented automatically if the fuel filter is changed in accordance with the instructions.

Note : QST30 engines with electric lift pumps can purge the system of air by energizing the electric lift pumps. Refer to Procedure 005-045 in Section 5. (</qs3/pubsys2/xml/en/procedures/57/57-005-045-tr.html>) Refer to Procedure 006-003 in Section 6. (</qs3/pubsys2/xml/en/procedures/57/57-006-003.html>)



Note : Manual venting for engines with mechanical lift pumps is required if any of the following is true.

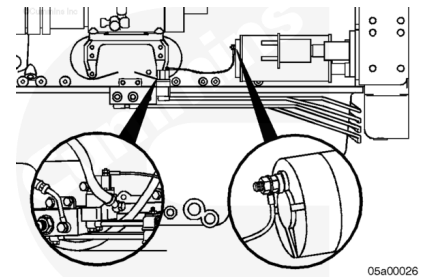
- Fuel filter is **not** filled prior to installation.
- Fuel injection pump is replaced.



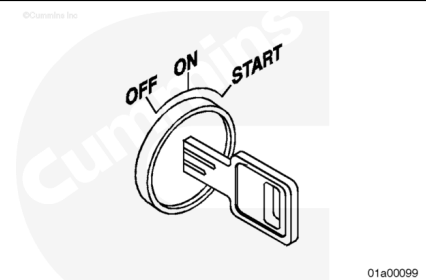
- High-pressure fuel line connections are loosened or fuel lines replaced.
- Initially starting the engine or starting up the engine after an extended period of time without engine operation.
- Vehicle fuel tank has been run until empty.

Prior to venting the generator drive engine, connect a 24-VDC wire from the positive terminal of the starter to the fuel shutoff solenoid post.

Note : Remove this wire before starting the engine.

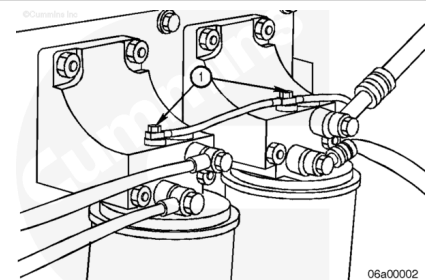


Prior to venting the industrial engine with mechanical lift pumps, place the keyswitch in the ON position, but do **not** crank the engine.



Loosen the relief valve (1) on the fuel filter head.

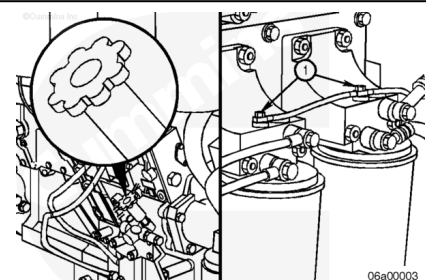
Note : On industrial engines with mechanical lift pumps, loosen the plug in the outlet port on the front of the fuel filter head.



Operate the plunger on the mechanical lift pump until the fuel flowing from the relief valve (1) (outlet fitting) is free of air.

Tighten the relief valve (1) (outlet fitting) on the fuel filters.

Torque Value: 11 n•m [97 in-lb]



Note : For generator/generator drive engines, the overflow valve is mounted on the face of the P8500 injection fuel pump. For industrial engines, the overflow line is on the EHAB. Refer to Procedure 005-066 in Section 5.

(/qs3/pubsys2/xml/en/procedures/57/57-005-066.html)

Loosen the overflow/pressure regulator fitting.

Operate the plunger on the mechanical fuel lift pump until the fuel flowing from the overflow/pressure regulator valve is free of air.

Tighten the overflow/pressure regulator valve. Refer to Procedure 005-066 in Section 5.

(/qs3/pubsys2/xml/en/procedures/57/57-005-066.html)

